

Population:

The population is the entire group of individuals or objects that we are interested in studying.

Example:

If we want to study the heights of all students in a college, all students in that college are the population.

Sample:

A sample is a smaller group selected from the population for the purpose of study.

Example:

If we select 100 students from that college and measure their heights, those 100 students are the sample.

Sampling Variation:

Sampling variation is the difference in results obtained from different samples selected from the same population.

For example, suppose a college has 10,000 students. If we select a sample of 100 students and calculate their average height, we may get 165 cm. If we select another 100 students, the average might be 167 cm.

Sampling Techniques

1. Probability Sampling

a) Simple Random Sampling

Every member has an equal chance of being selected.

Example: Selecting 50 students randomly from a list of 500 students.

b) Systematic Sampling

Members are selected at a fixed interval from a list.

Example: Selecting every 10th student from a list.

c) Stratified Sampling

The population is divided into different groups (strata), and samples are selected from each group.

Example: Selecting students separately from first year, second year, and third year.

d) Cluster Sampling

The population is divided into groups (clusters), and some clusters are selected randomly.

Example: Selecting 5 schools from a city and studying students in those schools.

2. Non-Probability Sampling

In non-probability sampling, every member of the population does not have a known or equal chance of being selected.

a) Convenience Sampling

People who are easily available are selected.

Example: Surveying students who are present in the classroom.

b) Purposive (Judgmental) Sampling

People are selected because they have a specific characteristic or knowledge needed for the study.

Example: Selecting experienced doctors for a medical study.

c) Quota Sampling

The population is divided into groups, and a fixed number (quota) is selected from each group.

Example: Selecting 50 males and 50 females for a survey.

d) Snowball Sampling

Existing participants help the researcher find other participants.

Example: Studying a hard-to-reach group by asking participants to refer others.

Sampling Bias

Sampling bias occurs when the sample selected for a study does not properly represent the population because some members have a greater or lesser chance of being selected.

As a result, the conclusions drawn from the sample may be incorrect or misleading.

Example:

Suppose a college wants to know whether students are satisfied with the college. If the researcher asks only students who are present in the library, the sample may not represent all students.

Estimate

An estimate is a value calculated from a sample that is used to approximate an unknown value of the population.

Example:

Suppose we want to know the average height of all students in a college.

We select 100 students. Their average height is 165 cm.

165 cm is an estimate of the average height of the entire population.

Stability

Stability refers to how consistent or reliable an estimate is when different samples are taken from the same population.

If repeated samples give similar results, the estimate is considered stable.

Example:

Suppose we take three samples and get average heights:

Sample 1 → 165 cm

Sample 2 → 166 cm

Sample 3 → 165 cm

The results are very similar, so the estimate is stable.

Central Limit Theorem (CLT)

The Central Limit Theorem states that when we take many random samples of sufficiently large size from a population, the sampling distribution of the sample means approximately normally distributed, even if the original population is not normally distributed.

Example:

Suppose the heights of students in a college are not normally distributed.

If we repeatedly take samples of 31 students and calculate the mean height of each sample, the different sample means will form a distribution that is approximately bell-shaped (normal).

Important points

It applies to sample means.

Larger sample sizes generally give a better approximation to a normal distribution.

The mean of the sampling distribution is approximately equal to the population mean

The standard deviation of the sampling distribution is called the standard error.

$$\mu_{\bar{x}} = \mu$$

$$SE = \sigma / \sqrt{n}$$

$$Z \text{ Score} = (\bar{x} - \mu) / SE$$

Where,

μ = Population mean

σ = Population standard deviation

n = Sample size

Suppose a university has 100,000 students.

We want to find the average height of all 100,000 students.

Instead of measuring every student, we take repeated random samples of 30 students.

For example:

Sample 1 → 31 students → mean height = 165 cm

Sample 2 → 31 students → mean height = 167 cm

Sample 3 → 31 students → mean height = 166 cm

Sample 4 → 31 students → mean height = 164 cm

Sample 5 → 31 students → mean height = 168 cm

And so on...

Each sample gives us one sample mean.

If we take a very large number of such samples, the collection of all these sample means forms the sampling distribution of the sample mean.

Point Estimate:

A point estimate is a single value obtained from a sample that is used to estimate the population parameter.

Suppose a university has 100,000 students.

We select a sample of 30 students and calculate their average height.

Suppose: Sample mean ($\bar{x}$) = 165 cm

Then:

Point estimate of population mean (μ) = $\bar{x}$ = 165 cm. So, 165 cm is the point estimate.

Confidence Interval:

A confidence interval (CI) gives a range of values within which the population parameter is estimated to lie.

The formula is:

Confidence Interval = $\bar{x} \pm z \times SE$

Where:

$\bar{x}$ = mean of the sample

z = z-value

SE = standard error

μ = population mean

For a 95% confidence interval:

$$z = 1.96$$

Therefore:

$$95\% \text{ CI} = \bar{x} \pm 1.96 \times \text{SE}$$

Example

Suppose:

Sample size = 30

Sample mean ($\bar{x}$) = 165 cm

Population SD (σ) = 10 cm

Confidence level = 95%

First calculate standard error:

$$\text{SE} = \sigma / \sqrt{n}$$

$$\text{SE} = 10 / \sqrt{30} \approx 1.83 \text{ cm}$$

Now calculate the confidence interval:

$$95\% \text{ CI} = \bar{x} \pm 1.96 \times \text{SE}$$

$$= 165 \pm 1.96 \times 1.83$$

$$= 165 \pm 3.59$$

Therefore:

$$95\% \text{ CI} = 161.41 \text{ cm to } 168.59 \text{ cm}$$

In simple words

Point estimate:

One value $\rightarrow$ 165 cm

Confidence interval:

A range $\rightarrow$ 161.41 to 168.59 cm

So, the sample mean ($\bar{x}$) gives us the point estimate, while the confidence interval gives us a range around that estimate.

Margin of Error

Margin of Error (MOE) tells us how far the confidence interval extends on either side of the point estimate.

In simple words:

Margin of error = the amount we add and subtract from the sample mean.

Formula

Margin of Error = $z \times SE$

Hypothesis Testing

Hypothesis testing is a statistical method used to determine whether there is enough evidence from a sample to support a claim about a population.

For example, suppose a university claims:

"The average height of our students is 165 cm."

We take a sample of students and use their data to test this claim.

In hypothesis testing, we make two hypotheses:

1. Null Hypothesis (H_0)

The null hypothesis states that there is no significant difference, no effect, or no relationship. It represents the existing or assumed situation.

For the university example:

$H_0: \mu = 165 \text{ cm}$

This means:

The population's average height is 165 cm.

2. Alternative Hypothesis (H_1 or H_a)

The alternative hypothesis states that there is a significant difference, effect, or relationship.

For the same example:

$H_1: \mu \neq 165 \text{ cm}$

This means:

The population's average height is different from 165 cm.

Simple way to remember

H_0 (Null) → No difference / No effect

H_1 (Alternative) → Difference / Effect

Important: We don't usually say " H_0 is proven true." We say either reject H_0 or fail to reject H_0 based on the evidence.

P-value

The p-value tells us how unusual our sample result would be if the null hypothesis (H_0) were actually true.

In simple words:

P-value = How much evidence do we have against H_0 ?

Small p-value → strong evidence against H_0

Large p-value → weak evidence against H_0

Statistical Significance

A result is called statistically significant when the p-value is smaller than the chosen significance level (α).

The most commonly used significance level is: $\alpha = 0.05$

So:

If $p < 0.05$ → statistically significant → Reject H_0

If $p \geq 0.05$ → not statistically significant → Fail to reject H_0

A/B Testing — Food Delivery App Example

Suppose a food delivery app wants to increase the number of users who place an order.

They want to test whether changing the "Order Now" button improves the order rate.

1. Objective

Goal: Increase the order conversion rate.

They create two versions:

Version A (Control): Existing "Order Now" button

Version B (Variant): New, redesigned "Order Now" button

2. Randomly divide users

Users are randomly divided into two groups:

Group A → Version A

Group B → Version B

For example, 20,000 users are included:

10,000 users see A

10,000 users see B

3. Collect the results

Suppose:

Version A:

10,000 users → 1,000 orders

Conversion rate:

$$1,000 / 10,000 = 10\%$$

Version B:

10,000 users → 1,200 orders

Conversion rate:

$$1,200 / 10,000 = 12\%$$

So B has a higher conversion rate.

4. Calculate the improvement

Absolute difference:

$$12\% - 10\% = 2 \text{ percentage points}$$

Relative improvement (lift):

$$(12\% - 10\%) / 10\% \times 100$$

$$= 20\% \text{ lift}$$

So Version B produced 20% more orders relative to A.

5. Hypothesis testing

We now ask:

Is the 2-percentage-point difference a real effect, or could it have happened simply because of random sampling variation?

Null hypothesis (H_0):

There is no difference between A and B.

$$H_0: p_A = p_B$$

Alternative hypothesis (H_1):

There is a difference between A and B.

$$H_1: p_A \neq p_B$$

6. Calculate the p-value

Suppose our statistical test gives:

$$p = 0.01$$

We use:

Significance level (α) = 0.05

Since:

$$p = 0.01 < 0.05$$

The result is statistically significant.

Therefore:

Reject H_0 .

There is sufficient evidence that the versions perform differently.

7. Business decision

If the experiment was designed correctly and the improvement is practically meaningful:

→ Food delivery app chooses Version B.

Complete A/B testing framework

Food delivery app

↓

Goal: Increase orders

↓

A = Old button

B = New button

↓

Randomly divide users

↓

Measure conversion rate

↓

A = 10%

B = 12%

↓

Calculate lift = 20%

↓

Hypothesis test → $p = 0.01$

↓

$p < 0.05$ → Statistically significant

↓

Choose B

The key idea

A/B testing is basically:

"We changed one thing → compared A and B → measured the difference → used statistics to determine whether the difference is likely real."